

PetroStream Ultra 2.0 - SBAA Basin Analysis Tool

An advanced petroleum geology dashboard for integrated source rock characterization, maturity analysis, and resource estimation of the Upper Devonian (Frasnian) interval in the SBAA Basin. Built with Streamlit and Plotly.

Figure 1: PetroStream Dashboard

Features

Geochemical Analysis

- **Rock-Eval Pyrolysis:** TOC, S2, Tmax, HI, OI analysis
- **Maturity Assessment:** Vitrinite reflectance (Ro) calculation using Jarvie (2007) proxy
- **Kerogen Typing:** Van Krevelen diagrams (HI vs OI)
- **Cross-Plots:** Interactive parameter correlation analysis

Spatial Visualization

- **3D Surface Mapping:** Interpolated property maps (Thickness, TOC, S2, Ro, HI)
- **Well Location Maps:** HC Potential classification with interactive hover data
- **Contour Analysis:** 2D contour generation with configurable resolution

Petrophysical Tools

- **Synthetic Log Viewer:** GR, RHOB, NPHI, RT, DT, PE tracks
- **Burial History Simulator:** 1D basin modeling with thermal evolution
- **Maturity Windows:** Oil generation, wet gas, dry gas zone identification

Resource Estimation

- **Volumetric Calculations:** OOIP/OGIP estimation using modified Jarvie (2012) methodology
- **Sensitivity Analysis:** Recovery factor vs drainage area scenarios
- **HC Generation Model:** Transformation ratio and expelled hydrocarbons

Reporting

- **Automated PDF Reports:** Technical documentation with basin statistics
- **CSV Export:** Raw data export for external analysis
- **Interactive HTML:** Dark-themed technical reports

Quick Start

Prerequisites

- Python 3.11 or higher
- pip package manager

Installation

```
# Clone the repository
git clone https://github.com/yourusername/petrostream.git
cd petrostream

# Install dependencies
pip install -r requirements.txt

# Run the application
streamlit run petrostream_app.py
```

Access the App

Open your browser and navigate to <http://localhost:8501>

Requirements

```
streamlit>=1.28.0
pandas>=2.0.0
numpy>=1.24.0
matplotlib>=3.7.0
scipy>=1.11.0
plotly>=5.17.0
openpyxl>=3.1.0
```

Architecture

```
petrostream/
├─ petrostream_app.py      # Main Streamlit application
├─ requirements.txt        # Python dependencies
├─ README.md               # This file
├─ data/
│   └─ sbaa_basin_data.csv # Well data (Rock-Eval, coordinates)
├─ assets/
│   ├── dashboard_preview.png # App screenshot
│   └─ logo.png              # PetroStream logo
└─ docs/
    ├── methodology.md       # Technical documentation
    └─ api_reference.md      # Code documentation
```

Methodology

Maturity Calculation

Vitrinite reflectance (Ro) is calculated from Tmax using the Jarvie (2007) proxy:

$$Ro = 0.018 \times Tmax - 7.16$$

Maturity Stage	Ro (%)	Tmax (°C)
Immature	< 0.6	< 435
Early Oil	0.6-0.9	435-445
Peak Oil	0.9-1.35	445-465
Wet Gas	1.35-2.0	465-480
Dry Gas	> 2.0	> 480

HC Potential Classification

Based on Peters & Cassa (1994) TOC thresholds:

Classification	TOC (%)
Poor	< 0.5
Fair	0.5-1.0
Good	1.0-2.0
Very Good	2.0-4.0
Excellent	> 4.0

Resource Estimation

Shale Oil-in-Place (OOIP):

$$OOIP = (Area \times Thickness \times \varphi \times (1-S_w)) / (5.615 \times Bo)$$

Gas-in-Place (OGIP):

$$OGIP = (Mass \times V_l \times P) / (P_l + P) / 10^9$$

Where:

- V_l = Langmuir volume (~100 scf/ton per % TOC)
- P_l = Langmuir pressure (500 psi)

🎨 UI/UX Design

- **Dark Luxury Theme:** Petroleum industry-inspired dark interface
- **Gold/Teal Color Scheme:** Professional geoscience visualization
- **Responsive Layout:** Optimized for desktop and tablet viewing
- **Interactive Plotly Charts:** Zoom, pan, hover tooltips
- **Real-time Calculations:** Instant parameter updates

📊 Data Sources

The application includes built-in data for 8 exploration wells in the SBAA Basin:

Well	X	Y	Thickness (m)	TOC (%)	S2	Tmax (°C)	HI	Q
SBAA-1	7.5	5.5	70	1.44	4.39	446	315	1
DECH-1	6.5	7.5	68	2.65	16.04	440	484	2

Table 3 – continued

Well	X	Y	Thickness (m)	TOC (%)	S2	Tmax (°C)	HI	
OTLA-1	1.8	1.2	54	1.49	6.91	437	467	3
BDW-1	6.2	6.1	48	0.57	2.04	460	404	2
ODZ-1	5.8	5.0	185	5.74	4.06	445	292	
OTRT-1	4.2	4.3	188	0.97	0.79	452	93	
LT-1bis	2.0	2.1	173	0.61	0.43	443	53	4
MGR-1	5.9	4.8	253	0.71	1.88	454	128	5

Development

Key Components

Data Loading & Caching:

```
@st.cache_data
def load_data():
    # Well data with calculated parameters (Ro, PI, HC_Potential, Maturity)
    ...
```

Theme Configuration:

```
def apply_theme(fig, title, height, ...):
    # Apply PetroStream dark theme to Plotly figures
    ...
```

Navigation System:

```
NAV_ICONS = {
    "$\vcenter{\hbox{\includegraphics[height=\ht\strutbox]{/usr/share/docminer/emoji/
    ↪ 1f3e0.png}}}}}$ Overview": "Overview",
    "$\vcenter{\hbox{\includegraphics[height=\ht\strutbox]{/usr/share/docminer/emoji/
    ↪ 1f4ca.png}}}}}$ Basin Registry": "Basin Registry",
    "$\vcenter{\hbox{\includegraphics[height=\ht\strutbox]{/usr/share/docminer/emoji/
    ↪ 1f9ea.png}}}}}$ Geochemical Lab": "Geochemical Lab",
    ...
}
menu = st.sidebar.radio("", list(NAV_ICONS.keys()))
```

Color Palette

Color Name	Hex Code	Usage
Gold Bright	#f0c040	Primary accent, headers
Gold Mid	#c8982a	Secondary elements
Teal	#00d4aa	Success, active states
Blue Data	#4499ff	Data visualization
Amber	#ff8c00	Warnings, gas windows
Red Warn	#ff4444	Errors, dry gas
BG Void	#04080f	Background

Table 4 – continued

Color Name	Hex Code	Usage
BG Panel	#0c1424	Cards, panels

Troubleshooting

Common Issues

ValueError in Plotly charts:

- Ensure all color values use 6-digit hex or rgba format (not 8-digit hex)
- Check that hovertemplate uses proper f-string formatting

Module not found errors:

```
pip install -r requirements.txt --upgrade
```

Streamlit cache issues:

```
streamlit cache clear
```

References

1. Jarvie, D.M. (2007) - Vitrinite reflectance proxy from Tmax
2. Peters, K.E. & Cassa, M.R. (1994) - Source rock evaluation
3. Jarvie, D.M. (2012) - Shale oil resource estimation
4. Pepper, A.S. & Corvi, P.J. (1995) - HC generation kinetics

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License

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Acknowledgments

- Streamlit team for the amazing framework
- Plotly team for interactive visualization tools
- Petroleum geochemistry community for methodology standards

 **PetroStream Ultra 2.0** - Advanced Petroleum Geology Dashboard